Name : S.Akilarasi
Reg no: 192512426

Ex No 8 : Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.

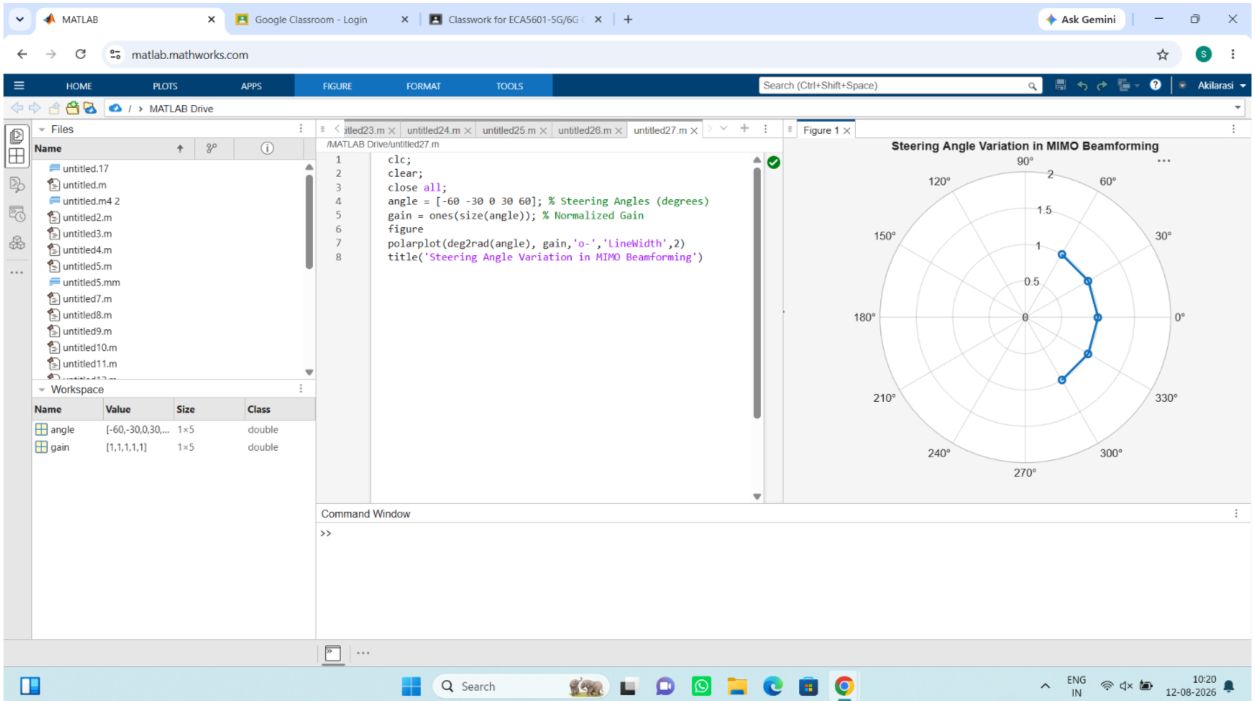
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.

3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Procedure:

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

Output:



Objective 2:

clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

bar(angle, gain)

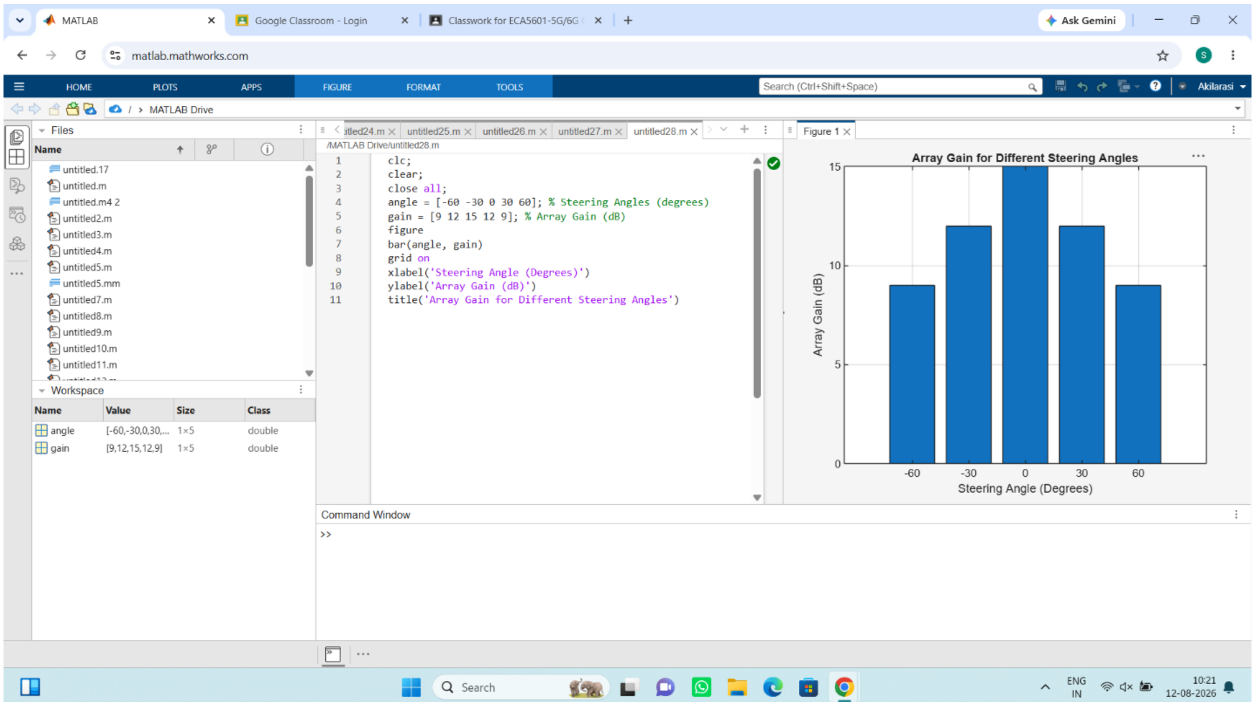
grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')

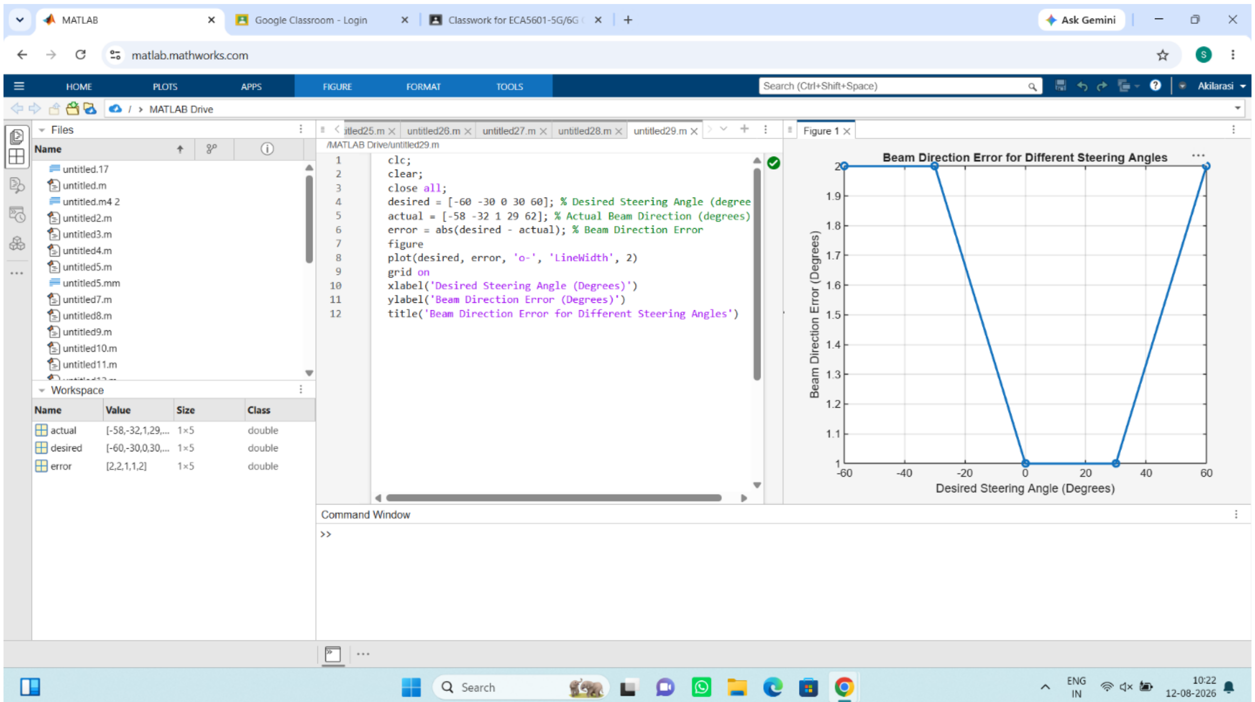
Output:



Objective 3:

```
clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)

actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')
```

Output:



Result

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.